

The distribution of beech leaf disease and the causal agents of beech bark disease (*Cryptococcus fagisuga*, *Neonectria faginata*, *N. ditissima*) in forests surrounding Lake Erie and future implications

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ABSTRACT

Over the past century, beech bark disease has dramatically altered the composition and structure of stands containing American beech (*Fagus grandifolia*). Management of beech bark disease has focused on reducing beech thickets and identifying resistant trees in aftermath forests. Beech leaf disease is a recently detected invasive disease, now also affecting beech forests. In 2019, a plot network was established in central North America to examine the extent of beech leaf disease spread and the severity of effects on trees and forests. Here, data from this plot network was used to determine the extent to which American beech is exposed to beech leaf disease, beech scale (*Cryptococcus fagisuga*), and beech bark disease (*C. fagisuga* and *Neonectria* spp. complex) in forests surrounding the Great Lakes in southwestern Ontario, northeastern Ohio, western Pennsylvania, and western New York. Beech leaf disease and scale were found to be well established, not only among sites but also in the different canopy layers. Beech bark disease was present, but occurrence was low. Pest accumulation differed between overstory and sapling layers. Beech leaf disease was the primary pest of saplings while both beech leaf disease and beech scale dominated in the overstory. Forest composition was similar throughout the study range; American beech, sugar maple, red maple, and white ash were abundant and common in the sapling and seedling layer. Few invasive plants were evident among saplings and seedlings. Sugar maple and red maple will likely fill small canopy gaps following beech decline or mortality. Intervention should be considered in stands with potential for increasing tree diversity or promoting beech leaf disease-resistant American beech trees to the overstory. New strategies are needed for beech conservation along with research to determine the extent of beech leaf disease resistance in American beech.

1. Introduction

American beech (*Fagus grandifolia* Ehrh.) is a common species, extending from Nova Scotia to Ontario and south to Florida and Texas (Burns and Honkala, 1990). American beech was once a dominant

species in the Great Lakes Region — about 3000–7000 years ago (Suffling et al., 2003; Cogbill, 2005). Since then, land clearing for agriculture and species migrations led to sugar maple (*Acer saccharum* Marshall), oak (*Quercus* spp.), and, in some areas, white pine (*Pinus strobus* L.) becoming more prevalent (Delcourt and Delcourt, 1987; Suffling et al.,

Abbreviations: BLD, beech leaf disease; BBD, beech bark disease.

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2003; Flinn et al., 2018). Although beech can form pure stands, it is more commonly found in mixed stands with other late successional species, including sugar maple (*Acer saccharum* Marsh.), yellow birch (*Betula alleghaniensis* Britt.), black cherry (*Prunus serotina* Ehrh.), and eastern hemlock (*Tsuga canadensis* (L.) Carrière) (Strobl and Bland, 2000).

A nut producing species, American beech masts are an important food source for 40 wildlife species including black bears, deer, birds, and small mammals (Jakubas et al., 2004; Storer et al., 2004; Bergeron et al., 2011). The high protein and fat content of the seeds are particularly important for black bear reproduction and survival (Jakubas et al., 2004). Beech wood is used for flooring, furniture, railway ties, pulp, and firewood (Burns and Honkala, 1990). Historically, beech was managed for mast and timber, especially in forests with few other mast-producing species.

In the 1890s, an exotic scale insect (*Cryptococcus fagisuga* Lindinger) arrived in Nova Scotia on imported European beech (*F. sylvatica* L.) seedlings (Houston, 1994). Commonly known as the beech scale, this insect spread to American beech and now occurs throughout much of the range of beech in North America (Garnas et al., 2011; Cale et al., 2017).

Beech bark disease (BBD), a canker disease complex involving the beech scale and two pathogenic fungi in the *Neonectria* genus, developed in forests infested with the beech scale. Beech scale feeding creates small fissures that predispose the host's bark tissue to subsequent invasion by one or more closely related canker fungi, *Neonectria ditissima* (Tul. & C. Tul.) Samuels & Rossman or *N. faginata* (M.L. Lohman, A.M.J. Watson & Ayers) Castl. & Rossman. *Neonectria ditissima* causes cankers on a variety of tree hosts while *N. faginata* has only been observed causing annual cankers on American beech trees following beech scale infestation (Castlebury et al., 2006; Stauder et al., 2020). *Neonectria ditissima* has been implicated in BBD both in North America and in Europe, whereas *N. faginata* appears to be restricted to beech in North America (Castlebury et al., 2006; Stauder et al., 2020). Beech bark disease has changed the structure and composition of eastern North American forests (Cale et al., 2017).

The detection of a new disease, called beech leaf disease (BLD), in the Great Lakes and Atlantic region is expected to result in further structural and compositional changes to North American beech stands. BLD was first detected in the Great Lakes region in 2012 and in the Atlantic region in 2019 (Ewing et al., 2019; Marra and LaMondia, 2020). Foliar symptoms included dark green, interveinal banding on infected leaves; a thickening of the leaf tissue; and leaf yellowing, necrosis, and curling resulting in thin tree crowns and tree mortality (Martin and Volk, 2021).

Monitoring and research revealed the presence of a nematode, *Litylenchus crenatae* ssp. *mccanni* (Carta et al. 2020) in infected leaves throughout the range of BLD (Reed et al., 2020). While this nematode must feed for BLD symptoms to occur, the potential role of additional pathogens, like bacteria, in symptom development remains under investigation (Burke et al., 2020; Carta et al., 2020). The North American population of the nematode displays differences in both morphology and host range from Japanese populations of *Litylenchus crenatae* Kanzaki, Ichihara, Aikawa, Ekino, and Masuya, 2019, thereby distinguishing it as its own subspecies, *L. crenatae* ssp. *mccannii* (Carta et al., 2020).

Mitigation of multiple forest pests, harmful insects and diseases, requires understanding their effects followed by devising or revising management regimes. Beech leaf disease and BBD are expected to affect forest stand composition and structure differently. BLD appears to disproportionately affect understory beech (Martin and Volk 2021) while BBD disproportionately affects overstory trees (Witter et al., 2005). During 2019, plots were established in BLD affected regions of Ontario (Canada), and Ohio, western Pennsylvania, and New York (USA). Here we describe the distribution of BLD, beech scale, and BBD in the U.S. and Ontario study areas, including the number of trees affected by each pest and severity of the infestation. Since early observations by researchers suggest these pests will not affect all size classes in the same

way, we focused on different canopy layers. We also characterized the sapling and seedling layer to determine which species are most likely to capitalize on the decline or death of American beech trees.

2. Material and methods

2.1. Study locations

In Ontario, 34 plots were established at 17 locations within 70 km of Lake Erie during August 2019 (Fig. 1). In the United States, 30 plots were established at 25 locations within 244 km of Lake Erie in Ohio, Pennsylvania, and New York between May and October 2019 (Fig. 1). Forests in these areas surrounding the Great Lakes have been referred to as Carolinian forest (Waldron, 2003), Central Hardwoods and Northern Hardwood – Conifer Forests (Fralish, 2003), and the Mixed Wood Plains and Atlantic Highlands (Wiken et al., 2011).

The strategy for plot placement in Ontario was to establish similar numbers of plots in areas with BLD, BBD, both diseases, and no disease. Historical monitoring data from Ontario's Ministry of Natural Resources and Forestry was consulted before requests were made to property owners in those areas. Requests to set up permanent plots were made to conservation authorities, provincial parks, land trusts, and private institutions. Forests selected in Ontario are largely classified as upland tolerant hardwood forests. Sugar maple represents 25% or more of the forest cover and either shares dominance or is associated with American beech, eastern hemlock (*Tsuga canadensis*, (L.) Carrière), white pine, red oak (*Quercus rubra* L.), white oak (*Quercus alba* L.), yellow birch, white birch (*Betula papyrifera* Marshall), white ash (*Fraxinus americana* L.), red maple (*Acer rubrum* L.), black maple (*Acer nigrum* F. Michx), black cherry, basswood (*Tilia americana* L.), American hop hornbeam (*Ostrya virginiana* (Mill.) K.Koch), bitternut hickory (*Carya cordiformis* (Wangenh.) K. Koch), shagbark hickory (*Carya ovata* (Mill.) K. Koch), and bitternut (*Juglans cinerea* L.). Tills and moraines are common in southern Ontario. Soils are acidic to slightly alkaline (pH 5.5–7.5) with hemlock and yellow birch favored on more alkaline sites (Strobl and Bland, 2000).

Presence of BLD was the primary factor when determining where to locate monitoring plots in the United States. Most plots were either placed in locations with well established BLD or near areas where BLD was newly discovered. Several plots were placed outside the known range of BLD to aid in its future detection. Plots were mostly located on public land including local parks, state parks, and state forests. Plots in the United States were generally set in high-quality, beech-maple reference forests in the western glaciated Allegheny Plateau or Erie-Ontario Lake Plain Section (Cleland et al., 2007; Hausman, 2015). Here, species associated with American beech commonly include sugar maple, red maple, green ash (*Fraxinus pennsylvanica* Marshall), white ash, black cherry, and tulip poplar (*Liriodendron tulipifera* L.).

2.2. Plot establishment

In Ontario, two plots were established in each forest. Each plot contained a minimum of three beech trees from the overstory (≥ 10 cm diameter at breast height, DBH) or the sapling layer (< 10 cm DBH, > 1.3 m tall). In the United States, one or two plots were established at each location. Each plot contained at least one beech tree.

Each permanent plot was composed of a 0.04 ha circular plot to assess overstory trees, a 0.004 ha subplot to assess saplings, and four 0.0003 ha subplots to assess seedlings. The sapling subplot was in the center of the overstory plot. Seedling subplots were established at the edge of the overstory plot in each cardinal direction.

Site characteristics, including slope (aspect, curvature, position) and landform shape, were determined for each plot. Overall basal area was calculated using a 10 basal area factor prism. Spherical crown densitometers were used to measure crown closure at the location of each seedling subplot in the four cardinal directions, 11.3 m from plot center.

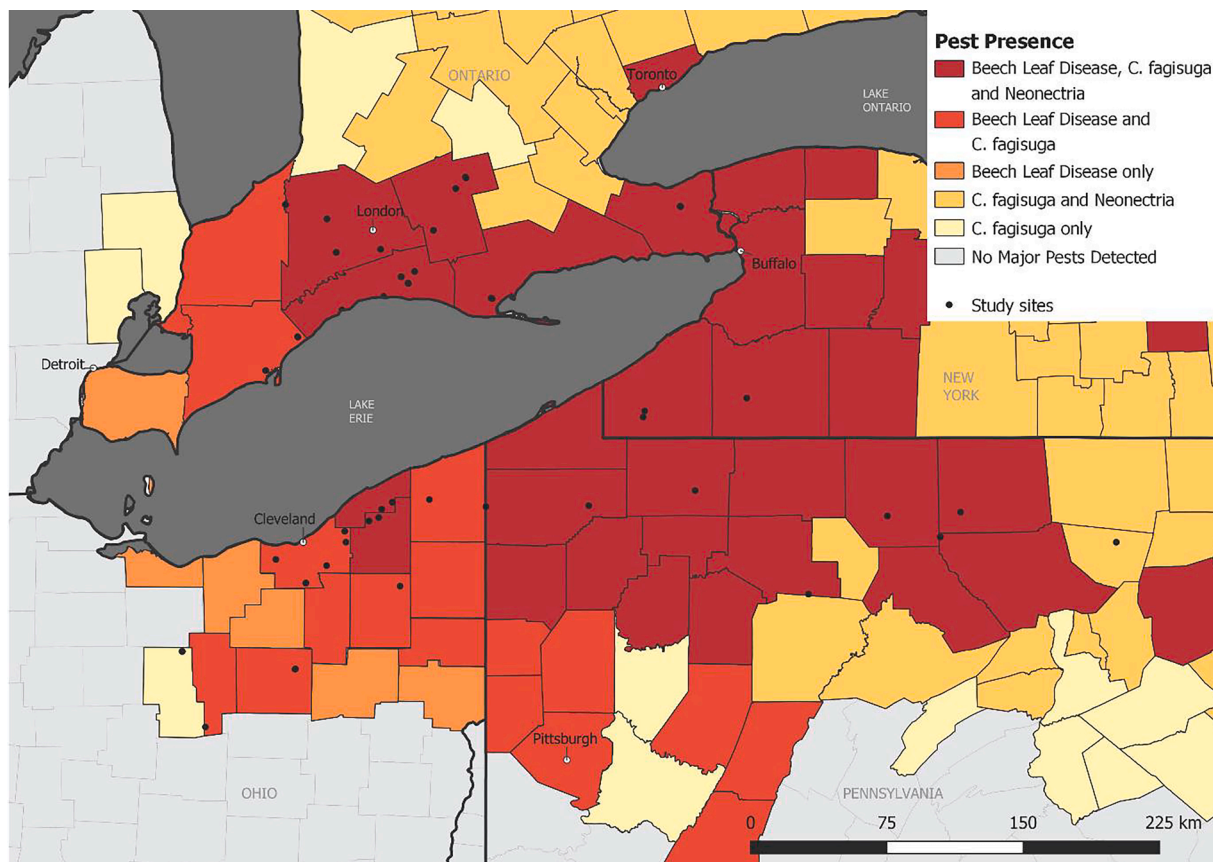


Fig. 1. County level distribution of pests as of October 2020 and locations of plots in the Great Lakes Region of Ontario, Canada and the United States. County level data was contributed by the Ontario Ministry of Natural Resources and Forestry, Biodiversity and Monitoring Section; USDA Forest Service, State and Private Forestry, Eastern Region; Michigan Department of Natural Resources; and Pennsylvania Department of Conservation and Natural Resources.

All overstory and sapling trees were identified to species and their DBH measured. Crown class was recorded for overstory trees as dominant, co-dominant, intermediate, or suppressed. For this study, the canopy layer was split into an upper and lower class with dominant and codominant trees in one class (D/C) and intermediate and suppressed trees (I/S) in the other. These trees differ in height, bark surface area, sun and wind exposure, all factors known to affect scale establishment and the severity of foliar diseases (Kimmins, 1997; Wargo, 1988; Agrios, 2005). All woody seedlings (<1.3 m tall) in the four subplots were counted and identified to species.

2.3. Pest assessment

Each plot took 1.5–4 h for a crew of three or more individuals to assess. Two or three surveyors spent approximately 10–15 min to assess overstory trees for BLD symptoms, scale and *Neonectria* cankers. For consistency, the same individuals from each crew, for each state and province, rated the same canopy layer.

The crown of each beech tree and sapling was visually assessed using binoculars to determine the presence and severity of BLD symptoms. Visual assessments included determining the percentages of leaves with no symptoms (normal), low severity symptoms (bands on <2/3 of leaf), and high severity symptoms (bands on >2/3 of leaf, chlorosis and tissue thickening often present). Percentage of symptom severity was divided into categories: 0%, 1–10%, 10–25%, 25–50%, 50–75%, and 75–100% of the crown. For seedlings, the presence of beech leaf disease was recorded as present or absent in each subplot.

Visual assessments with binoculars and hand lenses were made independently for beech scale and *Neonectria* (*N. faginata*, *N. ditissima*) at DBH on the north and south sides of each beech tree and sapling

(Wiggins et al., 2004); in addition, an overall score was assigned to the entire visible bole and root crown. Particular attention was given to the bole at DBH because *Neonectria* infections mostly occur on the lower bole (McLaughlin and Greifenhagen 2012). The categories for scale and *Neonectria* were: (0) none: no scale or *Neonectria*, (1) trace: 1–10 single scales or *Neonectria* colonies, (2) light: numerous single scales or *Neonectria* colonies scattered over the bole, (3) moderate: accumulation of scales or *Neonectria* colonies producing a clustered appearance, (4) moderate-heavy: clusters of scales or *Neonectria* colonies building to the point of beginning to appear to stream down the bole, (5) heavy: accumulation of scales or *Neonectria* on the bole increased to the point where portions of the bole appear white-washed or vertical accumulation lines produce the impression of scales or *Neonectria* streaming down the bole. Seedlings were not assessed for scale or *Neonectria*.

Neonectria was considered present when the following were observed: cankers with circle or lemon-shaped clusters of perithecia or eruption sites still visible from perithecia development. Trees with vertical patches of dead bark, cracked bark, or canker-like depressions on the bole were noted as suspect. When *Neonectria* was suspected but not visually present upon the initial assessment, sites were revisited at different times of the year to look for fruiting bodies.

2.4. Analysis

Data was analyzed using SAS 9.4 (SAS Institute, Cary NC) or R v3.5.1 (R Core Team, 2017). Fisher's exact test was used to compare the prevalence of BLD, scale infestation, and *Neonectria* among three beech size classes: (1) saplings, (2) suppressed trees, and (3) dominant trees. P-values from the individual tests were adjusted using the Holms method (Proc Multtest; step down Bonferonni). Fisher's exact test was also used

to compare pest prevalence between the Ontario and U.S. study area.

A generalized linear mixed model (PROC GLIMMIX) was used to compare the severity of BLD symptoms, beech scale infestations, and *Neonectria* infections among saplings, suppressed trees, and dominant trees from Ontario. Crown class was a fixed effect with each class having a random intercept. Before severity data was analyzed for BLD symptoms, the midpoint was calculated for each of the six categories for the percent canopy affected. The percentages of normal, low, and high severity leaves per tree were adjusted based on their relative contribution to the total (100%). BLD severity was determined by adding the proportion of low and high severity leaves together. Because BLD severity data are continuous from zero to one, a beta distribution with a logit-link was used for analyzing BLD severity data from Ontario and the United States. Categories of 1–5 were used to assess the severity of beech scale infestation and *Neonectria* infection. The Poisson distribution and log-link were specified when analyzing beech scale severity and a negative binomial distribution and log-link were specified when analyzing *Neonectria* severity. Tukey's HSD ($\alpha = 0.05$) was used to determine differences among means for all models.

The number of pests affecting beech saplings, suppressed trees, and dominant trees was also compared using GLMM with a log-link. The negative binomial distribution was specified for Ontario data while the Poisson distribution was specified for data from the United States. Beech

basal area was calculated using data from fixed area overstory and sapling plots.

3. Results

3.1. Ontario summary statistics

The survey included 646 live American beech trees, with 412, 85, and 149 trees in the sapling, I/S, and D/C classes, respectively. Twenty dead beech trees were recorded. The density of live beech stems ranged from 600 to 8900 stems ha^{-1} (mean = 3201 stems ha^{-1} ; Table S1). Basal area of live beech ranged from 5.3 to 39.0 $\text{m}^2 \text{ha}^{-1}$ (mean = 18.2 $\text{m}^2 \text{ha}^{-1}$). Basal area of dead beech ranged from 0 to 6.9 $\text{m}^2 \text{ha}^{-1}$ (mean = 0.3 $\text{m}^2 \text{ha}^{-1}$; Table S1).

For Ontario, American beech was the dominant species (>50% basal area) in 17 plots and a major component in 16 plots (20–50% basal area; Table S1). Other abundant species were sugar maple, red oak, red maple, basswood, hemlock, tulip poplar, and white elm (*Ulmus americana* L.). Beech was a minor component (10–20% basal area) in one plot dominated by sugar maple.

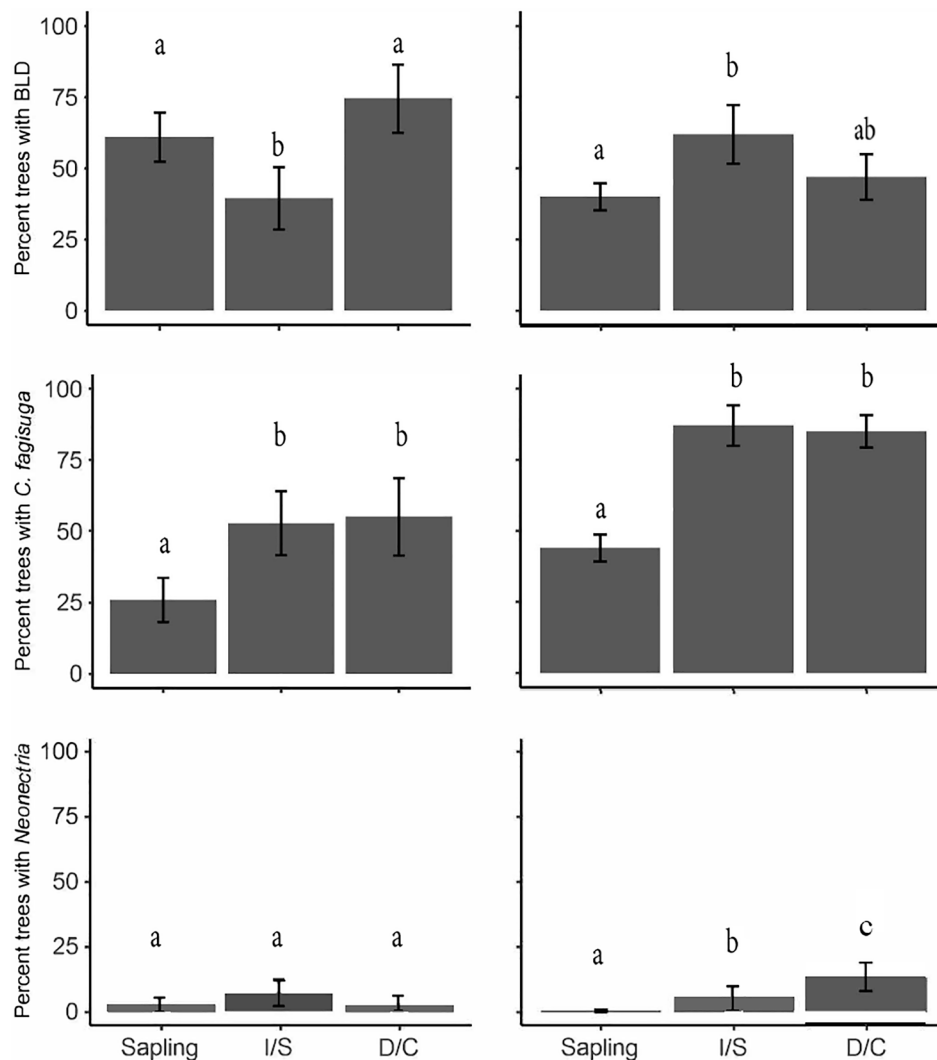


Fig. 2. Percentage of surveyed trees ($\pm 95\%$ C.I.) in each crown class with beech leaf disease, scale (*C. fagisuga*), and *Neonectria* (*N. faginata*, *N. ditissima*) in the United States (left) and Ontario (right) between May and August of 2019. Means with different letters are significantly different (Fishers Exact Test, Holm-Bonferroni method, $P \leq 0.05$). D/C – Dominant/Codominant, I/S – Intermediate/Suppressed.

3.2. Ontario disease assessment

Beech leaf disease was identified at 14 of 17 locations, with 25 of the 34 plots having trees with symptoms. The disease was present on 171 saplings, 53 I/S trees, and 70 D/C trees. BLD was 1.5 times more prevalent for I/S trees than saplings ($p < 0.001$; Fig. 2). BLD prevalence was not significantly different for I/S and D/C trees ($p > 0.05$). Beech leaf disease severity was 8% greater for I/S trees than D/C trees ($p = 0.002$; Fig. 3, Table 1). BLD severity for saplings was in between I/S and D/C trees and was not significantly different from either class ($p > 0.05$; Fig. 3, Table 1). Of the 208 beech seedlings assessed, 71 seedlings from 10 locations had BLD.

The scale, *C. fagisuga*, was present at all 34 plots. Scale was present on 185 saplings, 74 I/S, and 127 D/C trees. Scale was almost two times more prevalent on I/S ($p < 0.001$) and D/C trees ($p < 0.001$) than saplings (Fig. 2). Scale infestations were most severe on D/C trees followed by I/S trees and saplings ($p < 0.001$ for all comparisons; Fig. 3, Table 1).

Neonectria cankers (*N. faginata*, *N. ditissima*) were detected at seven locations, in nine of 34 plots. *Neonectria* was detected on one sapling, four I/S trees, and 20 D/C trees. It was 20 times less prevalent on saplings than on I/S trees ($p = 0.006$) and 2.9 times less prevalent on saplings than D/C trees ($p = 0.04$; Fig. 2). *Neonectria* was most severe on D/

Table 1

Hypotheses tests for significance of crown class by pest for beech leaf disease, *Cryptococcus fagisuga*, and *Neonectria* in the United States and Canada. Type III tests of fixed effects, GLIMMIX procedure.

Location, pest	Type III fixed effect	Test statistic	P-Value	Distribution
Canada				
Beech leaf disease	Crown class	$F_{2,638} = 6.11$	0.0024	Beta
<i>Cryptococcus fagisuga</i>	Crown class	$F_{2,638} = 143.31$	<0.0001	Poisson
<i>Neonectria</i>	Crown class	$F_{2,638} = 43.78$	<0.0001	Neg. Binomial
United States				
Beech leaf disease	Crown class	$F_{2,244} = 7.39$	0.0008	Beta
<i>Cryptococcus fagisuga</i>	Crown class	$F_{2,246} = 14.44$	<0.0001	Poisson
<i>Neonectria</i>	Crown class	$F_{2,246} = 3.40$	= 0.0349	Neg. Binomial

C trees followed by I/S trees ($p = 0.04$) and saplings ($p < 0.001$; Fig. 3, Table 1).

The number of pests varied by crown class ($F_{2,642} = 25.6$, $p < 0.001$). Saplings, with one pest on average, had less pests than D/C ($p < 0.001$)

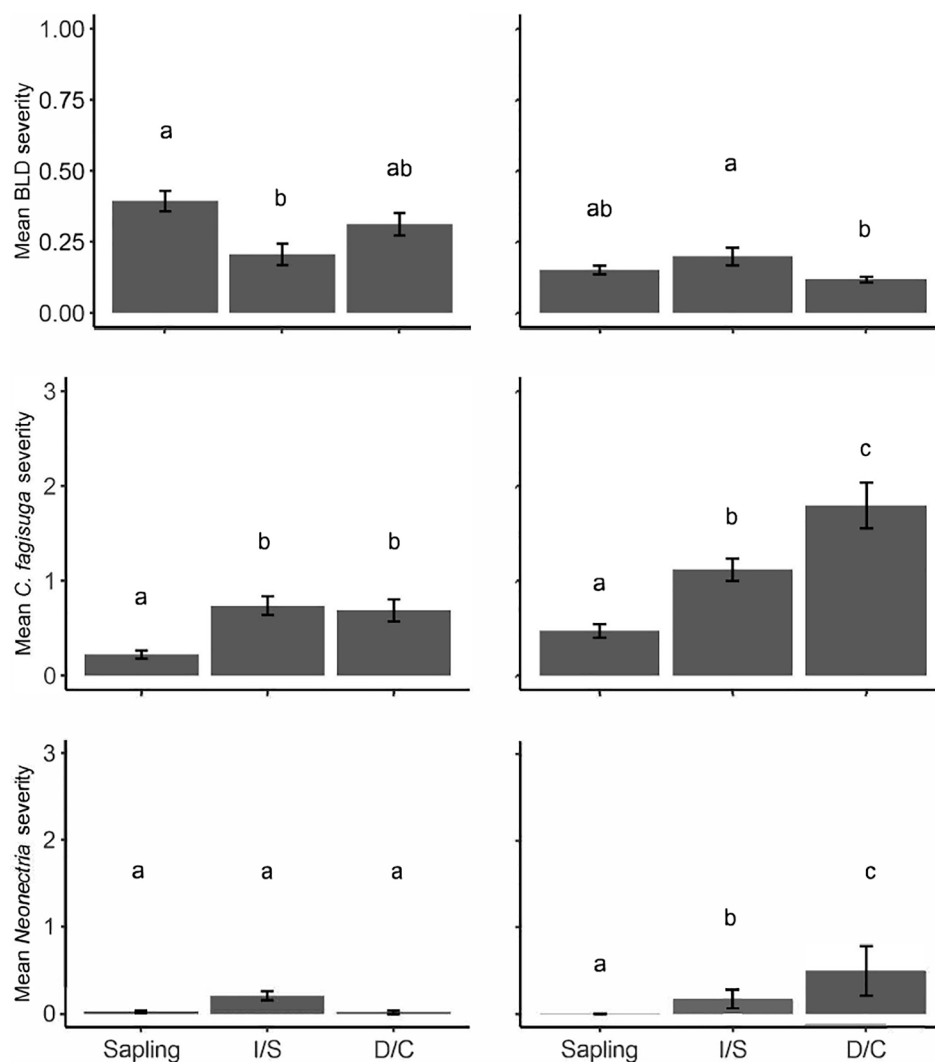


Fig. 3. Mean severity ($\pm$ SE) of beech leaf disease, beech scale (*C. fagisuga*), and *Neonectria* (*N. faginata*, *N. ditissima*) by crown class in the United States (left) and Ontario (right) between May and August 2019. Means with different letters are significantly different (Tukey's HSD, $P \leq 0.05$). D/C – Dominant/Codominant, I/S – Intermediate/Suppressed.

and I/S trees ($p = 0.004$, Fig. 4). Most saplings either had BLD or scale (Table 2). D/C and I/S trees averaged 1.5 pests ($p = 0.97$; Fig. 4). These trees typically had BLD and scale together or scale by itself (Table 2). Very few trees with scale or BLD had *Neonectria* canker and the disease was never observed by itself (Table 2). Fewer than one-third of saplings and one-sixth of D/C and I/S trees were pest free (Table 2).

3.3. Ontario – sapling and seedling plots

Twelve tree species and three shrub species were recorded in Ontario's sapling plots. American beech was the most abundant species, comprising 68% of all saplings recorded, followed by sugar maple (16%), white ash (5%), choke cherry (*Prunus virginiana* L.) (3%), spice bush (*Lindera benzoin* L.) (2%), American hop hornbeam (1%), and muscle wood (*Carpinus caroliniana* Walter) (1%) (Fig. 5). All other species were <1% of total stems recorded. American beech and sugar maple were the most common species, occurring in 100% and 62% of plots, respectively. The next most common species, white ash, was found in 18% of the plots.

Twenty-four tree species, 15 shrubs, and one vine were recorded in Ontario's seedling plots. Sugar maple represented 50% of all seedlings, followed by white ash (16%) and beech (7%; Fig. 5). Sugar maple, beech, and white ash were observed in 91%, 85%, and 79% of plots, respectively. Three invasive species were recorded: Japanese barberry (*Berberis thunbergii* DC.) (1 seedling), tatarian honeysuckle (*Lonicera tatarica* L.) (3 seedlings), and multiflora rose (*Rosa multiflora* Thunb.) (1 seedling).

3.4. United States summary statistics

The survey included 248 live American beech trees, with 123, 74, and 51 trees in the sapling, I/S, and D/C classes, respectively. The density of beech stems ranged from 40 to 882 stems ha^{-1} (mean = 334 stems ha^{-1} ; Table S2). Basal area for live beech ranged from 0 to 12.6 $\text{m}^2 \text{ha}^{-1}$ (mean = 4.4 $\text{m}^2 \text{ha}^{-1}$). Basal area for dead beech ranged from 0 to 6.5 $\text{m}^2 \text{ha}^{-1}$ (mean = 0.7 $\text{m}^2 \text{ha}^{-1}$).

For the United States, American beech was the dominant species in 11 plots, a major component in five plots, and a minor component in three plots (Table S2). In nine plots where beech was a trace component (<10% basal area), the dominant species were either red maple, red oak, or black cherry. Other species which were major components in plots included red maple, sugar maple, black cherry, cucumber tree (*Magnolia acuminata* L.), white oak, red oak, tulip tree, and bitternut hickory.

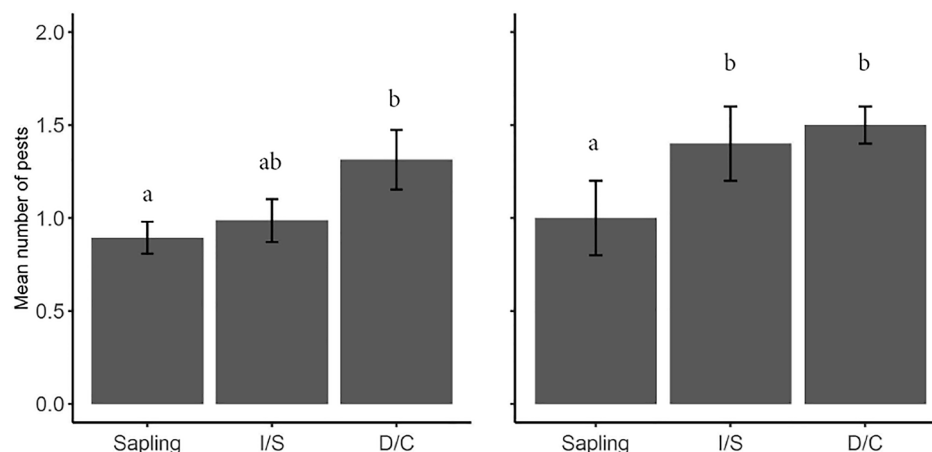


Fig. 4. Average number of pests per tree by size class ($\pm$ SE) for the United States (left) and Ontario (right) between May and August 2019. Means with different letters are significantly different (Tukey's HSD, $P < 0.05$). Pests are beech leaf disease, scale and *Neonectria* canker. D/C – Dominant/Codominant, I/S – Intermediate/Suppressed.

Table 2

Total number and percent of surveyed trees in Ontario and the United States that had beech leaf disease, *Cryptococcus fagisuga*, *Neonectria* or a combination of pests during 2019.

Location, pests detected	No. of trees (%) ^b			Total no. trees (%) ^b
	D/C ^c	I/S ^c	saplings	
Canada				
BLD, <i>C. fagisuga</i> , <i>Neonectria</i> ^a	6 (4)	1 (1)	1 (<1)	8 (1)
BLD, <i>C. fagisuga</i>	59 (40)	46 (54)	63 (15)	168 (26)
Scale, <i>Neonectria</i> ^a	14 (9)	3 (4)	0 (0)	17 (3)
BLD	0 (0)	0 (0)	106 (26)	106 (16)
<i>C. fagisuga</i>	48 (32)	24 (28)	121 (29)	193 (30)
No pests	22 (15)	11 (13)	121 (29)	154 (24)
United States				
BLD, <i>C. fagisuga</i> , <i>Neonectria</i> ^a	1 (2)	1 (1)	0 (0)	2 (1)
BLD, <i>C. fagisuga</i>	25(49)	24 (32)	20 (16)	69 (28)
Scale, <i>Neonectria</i> ^a	0 (0)	4 (5)	3 (2)	7 (3)
BLD	12 (24)	5 (7)	55 (45)	72 (29)
<i>C. fagisuga</i>	4 (8)	12 (16)	9 (7)	25 (10)
No pests	9 (18)	30 (39)	36 (29)	75 (30)

^a The combined activities of *C. fagisuga* and *Neonectria faginata* or *N. ditissima* result in beech bark disease.

^b total number of live trees surveyed in Ontario was 646 and in United States was 248.

^c D/C is Dominant/Codominant; I/S is Intermediate/Suppressed (Kimmins 1997).

3.5. United States disease assessment

Beech leaf disease was present in 17 of 30 plots and was found on 75 saplings, 30 I/S trees, and 38 D/C trees. BLD prevalence was similar between saplings and D/C trees ($p = 0.12$; Fig. 2). BLD was 1.5 times more prevalent for saplings than I/S trees ($p = 0.005$) and 1.9 times more prevalent for D/C trees than I/S trees ($p < 0.001$). BLD severity was twice as high for saplings compared to I/S trees ($p < 0.001$; Fig. 3; Table 1) but did not significantly differ between D/C and I/S trees ($p = 0.23$) or D/C trees and saplings ($p = 0.06$). Of the 82 beech seedlings assessed, 46 seedlings from 13 plots had BLD.

The scale, *C. fagisuga*, was present in 20 of 30 plots and was found on

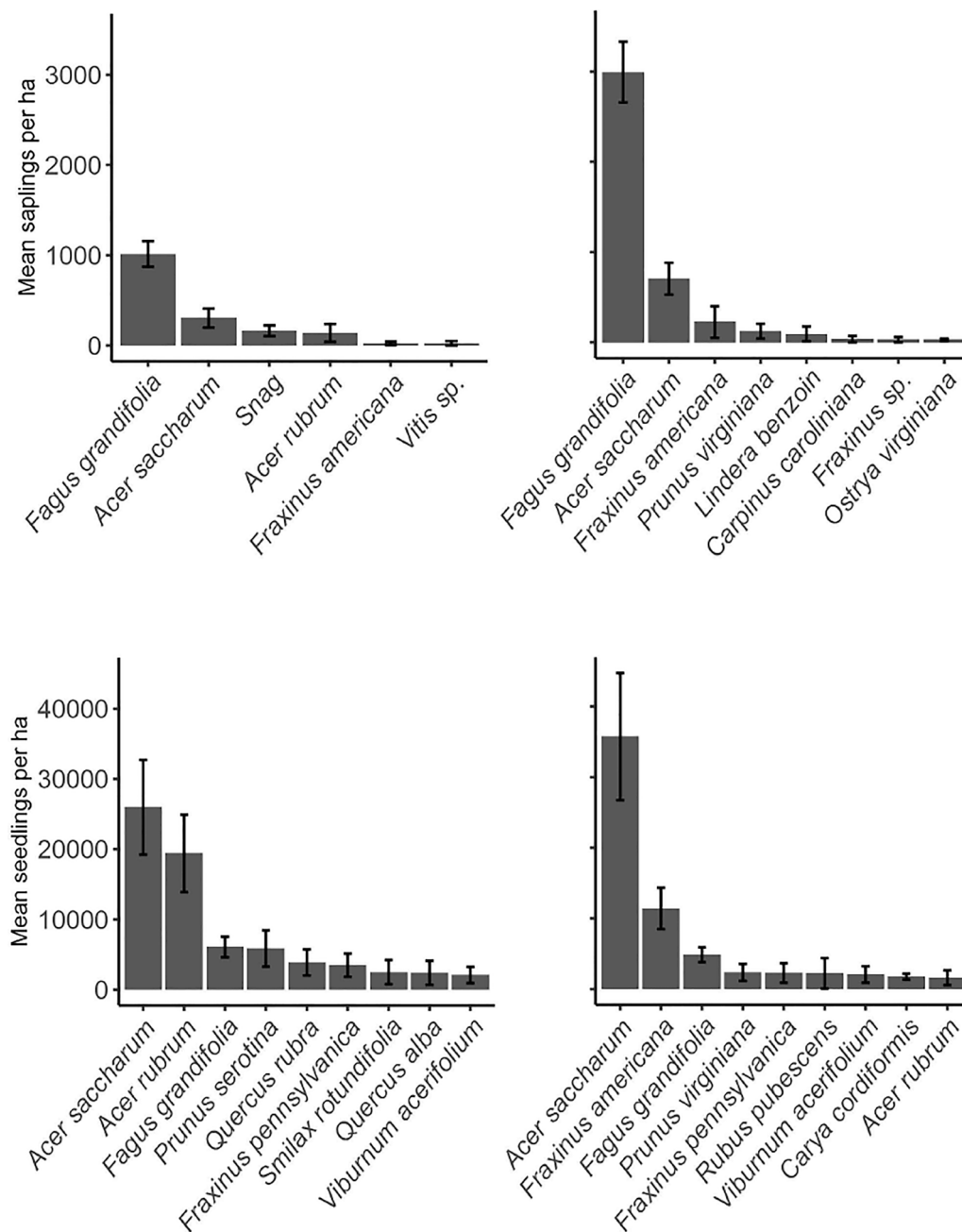


Fig. 5. Seedling and sapling regeneration rate per hectare ($\pm$ SE) in the United States (left) and Ontario (right) during 2019. Saplings with $>1\%$ and seedlings with $>2\%$ of the total count are displayed.

26 saplings, 40 I/S trees, and 28 D/C trees. Saplings were half as likely to have scale than I/S ($p < 0.0001$) or D/C trees ($p < 0.0001$; Fig. 2). Scale was significantly less severe for saplings than for suppressed ($p < 0.001$) and dominant trees ($p < 0.001$; Fig. 3, Table 1). Dominant and suppressed trees did not differ in scale severity ($p = 0.94$).

Neonectria cankers were present in 4 of 30 plots on three saplings, five I/S trees, and one D/C tree. The prevalence and severity of *Neonectria* was not significantly different among any size classes ($p > 0.05$; Fig. 2; Fig. 3; Table 1).

Overall pest count averaged 0.85 for saplings, 0.99 for I/S trees, and 1.31 for D/C trees (Fig. 4). The number of pests, BLD, scale, or *Neonectria*, varied by crown class ($F_{2,247} = 3.14$, $p = 0.045$). Saplings had significantly fewer pests than D/C trees ($p = 0.04$), but counts for I/S trees did not differ from either saplings ($p = 0.79$) or D/C trees ($p = 0.21$).

D/C trees most frequently had BLD and scale or BLD only, while I/S trees were most commonly pest free or had BLD and scale (Table 2). Most saplings had BLD or were pest free. *Neonectria* by itself or with other pests occurred least frequently for all canopy classes.

3.6. United States – sapling and seedling plots

The sapling plots had 14 tree and shrub species. Total count of sapling regeneration consisted primarily of beech (57%), sugar maple (17%), and red maple (8%; Fig. 5). Other species included white ash (1.4%) and *Vitis* spp. (1.4%). All other species represented $<1\%$ of total stems recorded. American beech, sugar maple, and red maple saplings were present in 93%, 33%, and 10% of plots, respectively.

There were 39 tree species, 9 shrubs and 3 vines in the seedling plots from the United States. Sugar maple comprised 53% of the total count of

seedlings, followed by red maple (21%), green ash (4%), black cherry (4%) and American beech (3%; Fig. 5). Sugar maple and beech seedlings were both present in 63% of plots while red maple was present in 90% of plots. Three invasive species were recorded: Japanese barberry (2 seedlings), buckthorn (*Frangula* sp.) (1 seedling), and multiflora rose (5 seedlings).

3.7. Comparison between study areas

BLD was more prevalent in the US compared to Ontario (54% vs 46% of trees; $p = 0.002$). Scale was more prevalent in Ontario compared to the US (60% vs 38% of trees; $p < 0.001$). Approximately 4% of trees were infected with *Neonectria* cankers in both study areas ($p > 0.05$).

4. Discussion

4.1. Beech scale and beech bark disease distribution and effects

Beech scale, *Neonectria* canker, and beech leaf disease each showed different levels of prevalence and severity across the landscape and across forest canopy layers. Beech scale was the most prevalent and widely distributed pest in the Ontario study area, with scale at all locations and 60% of study trees infested. Scale has existed in the Ontario study area since the 1960s when it was first detected in Elgin County (Keizer, 1998). The insect has been moved across Ontario by wind and human assisted transport and now occurs in all counties except Essex (MNRF 2016). Although scale has been in parts of the United States study area since 1984 (Cale et al., 2017), it was less prevalent than in Ontario where it was found in 67% of plots and on 38% of surveyed trees.

In both study areas, scale is disproportionately affecting overstory trees. In Ontario, scale infestation progressively increased with tree size. This difference was apparent in the United States as well but no significant difference was discernable between dominant-codominant and intermediate-suppressed trees. Scale infestations often start and become more severe on larger trees first as more surface area provides more opportunity for crawlers to land (Wargo, 1988).

Beech scale feeds on a range of tree sizes from large, overstory trees to trees as small as 1 cm diameter (Cale et al., 2017). Scale establishes on trees with rough bark and causes damage by using its stylet to puncture and feed on the phloem of young beech trees and the phelloderm of older trees (Ehrlich, 1934). Heavy feeding by the scale does not cause tree mortality but can reduce radial growth (Houston, 1994). Scale feeding elicits a host response resulting in higher concentrations of defense compounds in the bark of wounded trees (Cale et al., 2017). The biggest consequence of scale is the bark cracking and resulting *Neonectria* infections from feeding by dense clusters of scale (Cale et al., 2017). Pectinases released by scale insects also increase the bark's susceptibility to *Neonectria* infection (Houston, 1994).

Beech bark disease had the most limited distribution of the three pests in this study, with about 4% of beech trees infected in both study areas. The disease was first reported in the Ontario and U.S. study areas in the early 2000s and continues to develop slowly in this area relative to outlying areas to the north and east (Cale et al., 2017). To date no studies have determined the exact cause of slower BBD development near Lake Erie compared to other areas.

All sizes of beech trees are affected by BBD but it disproportionately affects large diameter trees (Witter et al., 2005). Having more surface area and bark with higher amino nitrogen content, large trees are more likely to be successfully colonized by large numbers of scale insects (Wargo, 1988). In our study, this was true for Ontario stands where incidence and severity were higher for larger trees. However, in the United States, incidence was highest for intermediate-suppressed trees and severity was similar among size classes; this finding is likely attributable to the absence of dominant and codominant beech in the few plots with BBD in the U.S. study area. Leak (2006) suggested that

BBD condition may not be related to tree size when beech is a small component in younger, more diverse stands.

The effects of BBD on forest composition, structure, and biodiversity are numerous and complex (Twery and Patterson, 1984; Lovett et al., 2006; Lovett et al., 2010; Garnas et al., 2011). In the killing front, trees with high density populations of scale become heavily infected with *Neonectria*, resulting in mortality of more than half of mature beech (Houston, 1994). *Neonectria* fungi grow into and kill inner bark and cambium creating both discrete and diffuse cankers. Coalescing cankers eventually girdle infected stems, resulting in tree mortality. This mortality typically occurs over a 10-year period (Houston, 1994). In this phase, BBD disproportionately kills larger beech. Thus, the arrival of BBD results in a shift towards smaller size classes (Garnas et al., 2011). Less than 1% of trees are estimated to be resistant to BBD (Houston, 1994).

Beech bark disease is the only forest disease in North America that can inadvertently perpetuate itself by increasing densities of its host to the detriment of other pest-free species (Cale et al., 2017). Increased light levels in stands brought about by disease-driven mortality or salvage often trigger prolific production of root sprouts, resulting in dense beech sapling understories that impede regeneration of less shade tolerant tree species (Houston, 1994; Morris et al., 2004; Beaudet and Messier, 2008; Koch, 2010; Cale et al., 2017). These clonal sprouts in aftermath forests are often genetically identical to their clonal parent, and are equally susceptible to BBD (Loo et al., 2005; Koch et al., 2010). Lichen and moss patches, *Xylococcus betulae* infestations, and roughening of bark with age result in these regenerating young beeches being colonized by scale and later *Neonectria* (Houston, 1994). The slow build up of cankers results in disfigurement and eventual death of most trees. Less than 1% of trees are estimated to be resistant to BBD (Houston, 1994). Thus, current stand-scale BBD management focuses on removal of understory beech to allow other tree species to establish (Forrester et al., 2003; Nelson and Wagner, 2011; Cale et al., 2017).

4.2. Beech leaf disease distribution and impact

Despite its much more recent detection, BLD has already become pervasive in forests surrounding Lake Erie in the United States and Ontario. Ewing et al. (2019) first suggested that BLD was rapidly spreading across the landscape. We found strong support for this rapid spread of symptomatic-BLD in our study area, with BLD symptoms already more widely distributed than BBD, a disease first detected decades prior (McLaughlin and Greifenhagen, 2012; Cale et al., 2017). The higher percentage of trees infected with BLD in the United States compared to Ontario supports the earlier arrival of BLD in the US versus Ontario.

Not only is BLD prevalent across the landscape, we found that it is affecting every canopy layer, dominant-codominant and intermediate-suppressed trees in the overstory as well as saplings and seedlings in the understory. In the United States, crowns of saplings were infected more often and had more severe symptoms than dominant and suppressed trees. This finding is consistent with a multiyear study being conducted in Cleveland Metroparks in northeast Ohio (Hausman and Volk, unpublished). In Ontario, a similar trend was observed but was not significant. This trend may become significant in Ontario as BLD becomes more severe in plots at the edge of the disease's current distribution.

Beech leaf disease affects understory saplings, with mortality observed within two to five years (Martin and Volk, 2021). Occasional mortality of overstory trees has been observed within seven years of infection; this pattern contrasts with that of BBD, which primarily affects dominant and suppressed trees. Leaf drop, bud abortion, and leaf curl in canopies of severely affected leaves allow more light to pass through to the understory (Volk et al., 2019). These changes in light levels are expected to alter plant communities on the forest floor. Stands affected by BLD or BLD and BBD will undergo compositional and structural shifts

as beech trees, especially seedlings and saplings, are increasingly affected.

The effects of multiple pests of beech in stands will be revealed by long term monitoring plots in Canada and the United States. Further attention should be paid to the combined influence of scale and beech leaf disease on overstory trees and the development of BBD in BLD-affected stands. Tree stress and nutrient availability are known to influence beech scale colonization and BBD development in European and American beech (Parker, 1975; Lonsdale, 1980; Wargo, 1988; Douglas, 2006; Cale et al., 2017).

4.3. Beech forest community change

Plots in both Canada and the United States contained a diverse array of tree species with sugar maple, red maple, American beech, and white ash the most prevalent. In the absence of active management, post-disturbance dynamics in mixed hardwood forests can influence and guide predictions of change in forests affected by one or more beech pests. As large and small tree morbidity and mortality progresses, saplings of sugar maple, red maple, and white ash can be expected to exploit canopy gaps of a variety of sizes (Canham, 1985; Burns and Honkala, 1990; Forrester et al., 2003). Sugar maple may benefit most as it grows rapidly in height and diameter, with even severely suppressed trees responding well to release (Strobl and Bland, 2000). White ash and red maple also grow quickly but both lose their shade tolerance with age and can be outcompeted by sugar maple when it is present. In this region, only ash surviving emerald ash borer would be expected to reach the canopy.

However, movement of maple and ash into the canopy is not guaranteed if BLD- or BBD resistant beech survive or if BLD fails to persist in stands. As is seen in many post-BBD sites, vigorous understory beech thickets can prevent maple progress to the canopy and prevent all but the most shade tolerant species from establishing (Cale et al., 2017). This is true not only for BBD, but other canopy release events: following a major ice storm, Beaudet et al. (2007) found that despite sugar maple's higher initial abundance and lesser shade tolerance, sugar maple sapling growth was not able to outcompete beech. Maple saplings are also preferentially browsed by deer; in some circumstances, herbivory can eliminate their competitive advantage, preventing ascent to the overstory (Beaudet et al., 2007; Bose et al., 2018).

The emergence of BLD is occurring in a forest landscape already crowded with influences, with invasive pests and land use history chief among them. Structural complexity and diversity, especially understory diversity, has been declining in eastern North American forests (Webster et al., 2018). Morbidity and mortality from these two diseases may ultimately advance the onset of uneven-aged conditions in second-growth hardwood forests, promoting younger trees into the canopy. During this process, loss of beech in both canopy and understory layers could provide an opportunity for shade intolerant species to establish in mixed hardwood forests that have undergone mesophication over the last century (Forrester et al., 2003; Kasson and Livingston, 2012; Radcliffe et al., 2020).

Invasive plants also have the potential to fill gaps left by declining beech. Increasing pest pressure on forests may reduce tree diversity and overall resilience (Boyd et al., 2013). In this study, beech plots had few invasive species. Maple-beech forests often have sparse understories due to low understory light levels (Strobl and Bland, 2000). Anticipated losses of beech from pests are expected to increase the abundance of invasive plants in forested areas, especially small woodlots and forests near urban areas. Buckthorn, some honeysuckles, barberry, privet, and autumn olive can take advantage of small canopy gaps, especially when soils are disturbed (Pearce, 1992). If active intervention is chosen to counteract the loss of beech, the risk of invasive plant establishment should be a management consideration.

4.4. Management and research needs for North American beech

Monitoring for compositional and structural change both in BLD affected forests and those where BLD will potentially spread is essential to understanding the ecological implications of this new disease. Examining the timing of BLD presence relative to tree age and size is also needed, as this may affect competitiveness of sprouting beech in the understory.

Some areas may lose significant numbers of beech where BLD and BBD are both present. While beech trees in BBD-affected areas are considered an undesirable timber species, they are an important wildlife species (Jakubas et al., 2004; Storer et al., 2004). Active management to promote shade intolerant species (oaks, hickories), especially those that support wildlife by mast production, may be necessary to mitigate the loss of hard mast from abundant beech trees. Promoting shade intolerant species would also increase diversity as many study area stands were almost entirely composed of mesophytic species.

Fortunately, an estimated 1% of American beech trees have some resistance to BBD (Houston, 1983; MacKenzie, 2005). Small numbers of beech with few or no BLD symptoms have been observed in BLD infected stands. Canopy gap size determines whether these trees ascend to the overstory (Burns and Honkala, 1990). In stands where overstory beech mortality or canopy thinning is likely to result in large canopy gaps (i.e., stands with high beech cover), active management may be needed to promote BLD resistant beech. In large gaps, faster growing, associated and intolerant species will overtop beech seedlings. These gaps also promote soil drying which disfavors beech seedlings and exposes them to sun scald (Strobl and Bland, 2000).

Increased support of beech breeding programs, with a BLD-resistance component, is urgently needed in the United States and Canada. Previously, land managers were encouraged to remove scale susceptible beech from aftermath forests so that the genetics of beech populations would shift towards resistant genotypes (Koch et al., 2010; McLaughlin and Greifenhagen, 2012). This idea of increasing genetic resistance by active management of natural regeneration is unlikely to succeed with BLD killing potentially scale resistant sprouts and saplings, greatly reducing overall genetic diversity. Breeding programs also need to factor in the beech leaf mining weevil, an unregulated invasive pest spreading across Nova Scotia. This pest could undermine efforts to place BLD-BBD resistant trees into forests (Sweeney et al., 2020). Options for preserving pest resistant beech in Canada are limited as the beech breeding program's future is uncertain due to a lack of funding. Cryopreserving beech embryos from Canada to develop a beech conservation collection, as is being done for butternut, may be one possible approach to support future research and restoration efforts (Pike et al., 2020).

5. Conclusion

This study demonstrates that BLD has rapidly spread in forests surrounding the Great Lakes and is affecting all size classes of beech. It has expanded more rapidly than BBD, which is no longer the sole concern for this area. Beech leaf disease is now the primary concern for beech seedlings and saplings while BLD and scale combined are the primary concern for overstory beech trees.

In these sugar maple-dominated forests, the percent cover of sugar maple is expected to increase due to its prevalence and abundance in sapling and seedling layers. This composition change may be a concern for forest managers interested in managing forests for conservation, sustainability, resilience, and ecosystem services such as mast for wildlife. As suggested by Webster et al. (2018), careful consideration of the site, available regeneration, herbivory levels, risk of introducing invasive plants, and future climate is needed when deciding if and how action is taken.

For BLD-affected areas, past suggestions of removing scale susceptible beech from stands to increase the proportion of scale resistant beech is unlikely to be effective. Instead, efforts may be better focused

on collecting genetic material from scale and BLD resistant beech from a wide geographic area and breeding genetically diverse BLD-BBD resistant beech.

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CRediT authorship contribution statement

Sharon E. Reed: Investigation, Writing – original draft, Conceptualization. **Daniel Volk:** Investigation, Writing – original draft. **Danielle K.H. Martin:** Methodology, Conceptualization, Investigation. **Constance E. Hausman:** Methodology, Investigation, Conceptualization. **Tom Macy:** Methodology, Investigation. **Tim Tomon:** Methodology, Investigation. **Stella Cousins:** Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- Agrios, G., 2005. *Plant Pathology*, fifth ed. Elsevier Academic Press, Burlington MA, USA.
- Beaudet, M., Brisson, J., Gravel, D., Messier, C., 2007. Effect of a major canopy disturbance on the coexistence of *Acer saccharum* and *Fagus grandifolia* in the understorey of an old-growth forest. *J. Ecol.* 95, 458–467. <https://doi.org/10.1111/j.1365-2745.2007.01219.x>.
- Beaudet, M., Messier, C., 2008. Beech regeneration of seed and root sucker origin: a comparison of morphology, growth, survival, and response to defoliation. *For. Ecol. Manag.* 255, 3659–3666. <https://doi.org/10.1016/j.foreco.2008.03.010>.
- Bergeron, P., Réale, D., Humphries, M.M., Garant, D., 2011. Anticipation and tracking of pulsed resources drive population dynamics in eastern chipmunks. *Ecology* 92, 2027–2034. <https://doi.org/10.1890/11-0766.1>.
- Bose, A.K., Wagner, R.G., Roth, B.E., Weiskittel, A.R., 2018. Influence of browsing damage and overstory cover on regeneration of American beech and sugar maple nine years following understory herbicide release in central Maine. *New For.* 49, 67–85. <https://doi.org/10.1007/s11056-017-9605-5>.
- Boyd, I.L., Freer-Smith, P.H., Gilligan, C.A., Godfray, H.C.J., 2013. The consequence of tree pests and diseases for ecosystem services. *Science* 342, 1235773. <https://doi.org/10.1126/science.1235773>.
- Burke, D.J., Hoke, A.J., Koch, J., 2020. The emergence of beech leaf disease in Ohio: probing the plant microbiome in search of the cause. *For. Pathol.* e12579 <https://doi.org/10.1111/efp.12579>.
- Burns, R.M., Honkala, B.H., 1990. *Silvics of North America: 2. Hardwoods*. U.S. Department of Agriculture, Forest Service, Washington, DC, p. 877.
- Cale, J.A., Garrison-Johnston, M.T., Teale, S.A., Castello, J.D., 2017. Beech bark disease in North America: over a century of research revisited. *For. Ecol. Manag.* 394, 86–103. <https://doi.org/10.1016/j.foreco.2017.03.031>.
- Canham, C.D., 1985. Suppression and release during canopy recruitment in *Acer saccharum*. *Bull. Torrey Bot. Club* 112, 134–145. <https://doi.org/10.2307/2996410>.
- Carta, L.K., Handoo, Z.A., Li, S., Kantor, M., Baughan, G., McCann, D., Gabriel, C.K., Yu, Q., Reed, S., Koch, J., Martin, D., Burke, D.J., 2020. Beech leaf disease symptoms caused by newly recognized nematode subspecies *Litylenchus crenatae maccanii* (Anguinata) described from *Fagus grandifolia* in North America. *For. Pathol.* e12580. <https://doi.org/10.1111/efp.12580>.
- Castlebury, L.A., Rossman, A.Y., Hyten, A.S., 2006. Phylogenetic relationships of *Neonectria/Cylindrocarpum* on *Fagus* in North America. *Can. J. Bot.* 84, 1417–1433. <https://doi.org/10.1139/b06-105>.
- Cleland, D.T., Freeouf, J.A., Keys, J.E., Nowacki, G.J., Carpenter, C.A., McNab, W.H., 2007. Ecological Subregions: Sections and Subsections for the conterminous United States. In: Sloan, A.M. (Ed.), Gen. Tech. Report WO-76D [Map on CD-ROM]. U.S. Department of Agriculture, Forest Service, Washington, DC, pp. presentation scale 1: 3,500,000; colored.
- Cogbill, C.V., 2005. Historical biogeography of capitalized beech. In: Evans, C.A., Lucas, J.A., Twery, M.J. (Eds.), *Beech Bark Disease: Proceedings of the Beech Bark Disease Symposium*. Gen. Tech. Rep. NE-331. Newton Square PA, US. Department of Agriculture Forest Service, Northern Research, Station, pp. 16–24. <https://doi.org/10.2737/NE-GTR-331>.
- Delcourt, P.A., Delcourt, H.R., 1987. Late-Quaternary dynamics of temperate forests: applications of paleoecology to issues of global environmental change. *Quat. Sci. Rev.* 6, 129–146. [https://doi.org/10.1016/0277-3791\(87\)90030-8](https://doi.org/10.1016/0277-3791(87)90030-8).
- Douglas, A.E., 2006. Phloem-sap feeding by animals: problems and solutions. *J. Exp. Bot.* 57, 747–754.
- Ehrlich, J., 1934. The beech bark disease: a *Nectria* disease of *Fagus*, following *Cryptococcus fagi* (Bair.). *Can. J. Res.* 10, 593–692.
- Ewing, C.J., Hausman, C.E., Pogacnik, J., Slot, J., Bonello, P., 2019. Beech leaf disease: an emerging forest epidemic. *For. Pathol.* 49, e12488. <https://doi.org/10.1111/efp.12488>.
- Flinn, K.M., Mahany, T.P., Hausman, C.E., 2018. From forest to city: plant community change in northeast Ohio from 1800 to 2014. *J. Veg. Sci.* 29, 297–306. <https://doi.org/10.1111/jvs.12621>.
- Forrester, J.A., McGee, G.G., Mitchell, M.J., 2003. Effects of beech bark disease on aboveground biomass and species composition in a mature northern hardwood forest, 1985 to 2000. *J. Torrey Bot. Soc.* 130, 70–78. <https://doi.org/10.2307/3557531>.
- Fralish, J.S., 2003. The central hardwood forest: its boundaries and physiographic provinces. In: Van Sambeek, J.W., Dawson, J.O., Ponder Jr., F., Loewenstein, E.F., Fralish, J.S. (Eds.), *Proceedings of the 13th Central Hardwood Forest Conference*. U. S. Department of Agriculture, Forest Service, North Central Research Station, pp. 1–20.
- Garnas, J.R., Ayres, M.P., Liebhold, A.M., Evans, C., 2011. Subcontinental impacts of an invasive tree disease on forest structure and dynamics. *J. Ecol.* 99, 532–541. <https://doi.org/10.1111/j.1365-2745.2010.01791.x>.
- Hausman, C.E., 2015. A regional ecosystem monitoring and assessment program for the Lake Erie-Allegheny Plateau Region. In: Cleveland Metroparks Technical Report 2015/NR-5. Cleveland Metroparks, Parma, Ohio.
- Houston, D.R., 1983. American beech resistance to *Cryptococcus fagisuga*. In: IUFRO Beech Bark Disease Working Party Conference. U.S. Forest Service, Wen, CT USA, pp. 3841.
- Houston, D.R., 1994. Major new tree disease epidemics: beech bark disease. *Ann. Rev. Phytopathol.* 32, 75–87. <https://doi.org/10.1146/annurev.py.32.090194.000451>.
- Jakubas, W.J., McLaughlin, C.R., Jensen, P.G., McNulty, S.A., 2004. Alternate year beechnut production and its influence on bear and marten populations. In: Evans, C.A., Lucas, J.A., Twery, M.J. (Eds.), *Beech Bark Disease: Proceedings of the Beech Bark Disease Symposium*. Gen. Tech. Rep. NE-331. Newton Square PA, US. Department of Agriculture Forest Service, Northern Research Station, pp. 79–87. <https://doi.org/10.2737/NE-GTR-331>.
- Kasson, M.T., Livingston, W.H., 2012. Relationships among beech bark disease, climate, radial growth response and mortality of American beech in northern Maine. *USA. For. Pathol.* 42, 199–212. <https://doi.org/10.1111/j.1439-0329.2011.00742.x>.
- Keizer, A.J., 1998. Results of forest insect and disease surveys in the eastern region of Ontario, 1987. *Forest Districts: Napanee, Tweed, Carleton Place, Cornwall and Brockville*. In: Great Lakes Forestry Centre, Canadian Forestry Service, Government of Canada, Sault Ste Marie, Ontario, pp. 47.
- Kimmins, J.P., 1997. *Forest Ecology: A Foundation for Sustainable Management*. Prentice Hall.
- Koch, J.L., 2010. Beech bark disease: the oldest “new” threat to American beech in the United States. *Outlooks Pest Manag.* 21, 64–68. <https://doi.org/10.1564/21apr03>.
- Koch, J.L., Carey, D.W., Mason, M.E., Nelson, C.D., 2010. Assessment of beech scale resistance in full- and half-sibling American beech families. *Can. J. For. Res.* 40, 265–272. <https://doi.org/10.1139/X09-189>.
- Leak, W.B., 2006. Fifty-year impacts of the beech bark disease in the Bartlett Experimental Forest, New Hampshire. *North. J. Appl. For.* 23, 141–143.
- Lonsdale, D., 1980. *Nectria* infection of beech bark in relation to infestation by *Cryptococcus fagisuga* Lindiger. *Eur. J. For. Pathol.* 10, 161–168. <https://doi.org/10.1111/j.1439-0329.1980.tb00022.x>.
- Loo, J., Ramirez, M., Krasowski, M., 2005. American beech vegetative propagation and genetic diversity. In: Evans, C.A., Lucas, J.A., Twery, M.J. (Eds.), *Beech Bark Disease: Proceedings of the Beech Bark Disease Symposium*. Gen. Tech. Rep. NE-331. Newton Square PA, US. Department of Agriculture Forest Service, Northern Research Station, pp. 106–112. <https://doi.org/10.2737/NE-GTR-331>.
- Lovett, G.M., Arthur, M.A., Weathers, K.C., Griffin, J.M., 2010. Long-term changes in forest carbon and nitrogen cycling caused by an introduced pest/pathogen complex. *Ecosystems* 13, 1188–1200. <https://doi.org/10.1007/s10021-010-9381-y>.

- Lovett, G.M., Canham, C.D., Arthur, M.A., Weathers, K.C., Fitzhugh, R.D., 2006. Forest ecosystem responses to exotic pests and pathogens in eastern North America. *Bioscience* 56, 395–405. [https://doi.org/10.1641/0006-3568\(2006\)056\[0395:Fertep\]2.0.CO;2](https://doi.org/10.1641/0006-3568(2006)056[0395:Fertep]2.0.CO;2).
- MacKenzie, M., 2005. Survival of the fittest: beech bark disease-resistant beech will leave more offspring. In: Evans, C.A., Lucas, J.A., Twery, M.J. (Eds.), *Beech Bark Disease: Proceedings of the Beech Bark Disease Symposium*. Gen. Tech. Rep. NE-331. Newton Square PA, US. Department of Agriculture Forest Service, Northern Research Station, pp. 65–67. <https://doi.org/10.2737/NE-GTR-331>.
- Marra, R.E., LaMondia, J.A., 2020. First report of beech leaf disease, caused by the foliar nematode, *Litylenchus crenatae mccannii*, on American beech (*Fagus grandifolia*) in Connecticut. *Plant Dis.* 104, 2527. <https://doi.org/10.1094/pdis-02-20-0442-pdn>.
- Martin, D., Volk, D., 2021. Pest Alert: Beech leaf disease. USDA Forest Service, Eastern Region. State and Private Forestry, Milwaukee, WI.
- McLaughlin, J.A., Greifenhagen, S., 2012. Beech bark disease in Ontario: a primer and management recommendations. In: Forest Research Note 71. Ontario Ministry of Natural Resources, Ontario Forest Research Institute, Sault Ste. Marie, ON, Canada.
- MNRF, 2016. Survival and reproductive capability of the beech scale (*Cryptococcus fagisuga*) and the beech bark disease fungus (*Neonectria faginata*) on beech firewood in Ontario. Ontario Ministry of Natural Resources, Science and Research Branch Technical Note TN-002, Sault Ste. Marie, ON. 8p. Canada. <https://doi.org/10.13140/RG.2.2.31490.89282>.
- Morris, A.B., Small, R.L., Cruzan, M.B., 2004. Variation in frequency of clonal reproduction among populations of *Fagus grandifolia* Ehrh. in response to disturbance. *Castanea* 69, 38–51. [https://doi.org/10.2179/0008-7475\(2004\)069<0038:VIFOCR>2.0.CO;2](https://doi.org/10.2179/0008-7475(2004)069<0038:VIFOCR>2.0.CO;2).
- Nelson, A.S., Wagner, R.G., 2011. Improving the composition of beech-dominated northern hardwood understories in northern Maine. *North. J. Appl. For.* 28, 186–193. <https://doi.org/10.1093/njaf/28.4.186>.
- Parker, E.J., 1975. Some investigations with Beech bark disease *Nectria* in southern England. *Eur. J. For. Pathol.* 5, 118–124. <https://doi.org/10.1111/j.1439-0329.1975.tb00453.x>.
- Pearce, C.M., 1992. Pattern analysis of forest cover in southwestern Ontario. *East Lakes Geogr.* 27, 65–76.
- Pike, C.C., Williams, M., Brennan, A., Woeste, K., Jacobs, J., Hoban, S., Moore, M., Romero-Severson, J., 2020. Save our species: a blueprint for restoring butternut (*Juglans cinerea*) across Eastern North America. *J. For.* 119, 196–206. <https://doi.org/10.1093/jofore/fvaa053>.
- R Core Team, 2017. A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria <https://www.R-project.org>.
- Radcliffe, D.C., Matthews, S.N., Hix, D.M., 2020. Beyond oak regeneration: modelling mesophytic sapling density drivers along topographic, edaphic, and stand-structural gradients in mature oak-dominated forests of Appalachian Ohio. *Can. J. For. Res.* 50, 1215–1227. <https://doi.org/10.1139/cjfr-2020-0111>.
- Reed, S.E., Greifenhagen, S., Yu, Q., Hoke, A., Burke, D.J., Carta, L.K., Handoo, Z.A., Kantor, M.R., Koch, J., 2020. Foliar nematode, *Litylenchus crenatae* ssp. *mccannii*, population dynamics in leaves and buds of beech leaf disease-affected trees in Canada and the US. *For. Pathol.* 50, e12599. <https://doi.org/10.1111/efp.12599>.
- Stauder, C.M., Garnas, J.R., Morrison, E.W., Salgado-Salazar, C., Kasson, M.T., 2020. Characterization of mating type genes in heterothallic *Neonectria* species, with emphasis on *N. coccinea*, *N. ditissima*, and *N. faginata*. *Mycologia* 112, 880–894. <https://doi.org/10.1080/00275514.2020.1797371>.
- Storer, A.J., Rosemier, J.N., Beachy, B.L., Flaspohler, D.J., 2004. Potential effects of beech bark disease and decline in beech abundance on birds and small mammals. In: Evans, C.A., Lucas, J.A., Twery, M.J. (Eds.), *Beech Bark Disease: Proceedings of the Beech Bark Disease Symposium*. Gen. Tech. Rep. NE-331. Newton Square PA, US. Department of Agriculture Forest Service, Northern Research Station, pp. 72–78. <https://doi.org/10.2737/NE-GTR-331>.
- Strobl, S., Bland, D., 2000. A silvicultural guide to managing southern Ontario forests. In: PaperWorks, Havelock, ON., pp. 654.
- Suffling, R., Evans, M., Perera, A., 2003. Presettlement forest in southern Ontario: ecosystems measured through a cultural prism. *For. Chron.* 79, 485–501. <https://doi.org/10.5558/tfc79485-3>.
- Sweeney, J.D., Hughes, C., Zhang, H., Hillier, N.K., Morrison, A., Johns, R., 2020. Impact of the invasive beech leaf-mining weevil, *Orchestes fagi*, on American beech in Nova Scotia, Canada. *Front. For. Glob. Chang.* 3 <https://doi.org/10.3389/ffgc.2020.00046>.
- Twery, M.J., Patterson, W.A., 1984. Variations in beech bark disease and its effect on species composition and structure of northern hardwood stands in central New England. *Can. J. For. Res.* 14, 565–574.
- Volk, D.R., Hausman, C.E., Martin, D., 2019. Beech leaf disease seasonal progression. In: Cleveland Metroparks Technical Report 2019/NR-07, Parma, Ohio.
- Waldron, G., 2003. *Trees of the Carolinian Forest*. Boston Mills Press, Erin, ON Canada.
- Wargo, P.M., 1988. Amino nitrogen and phenolic constituents of bark of American beech, *Fagus grandifolia*, and infestation by beech scale, *Cryptococcus fagisuga*. *Eur. J. For. Pathol.* 18, 279–290. <https://doi.org/10.1111/j.1439-0329.1988.tb00215.x>.
- Webster, C.R., Dickinson, Y.L., Burton, J.L., Frelich, L.E., Jenkins, M.A., Kern, C.C., Raymond, P., Saunders, M.R., Walters, M.B., Willis, J.L., 2018. Promoting and maintaining diversity in contemporary hardwood forests: Confronting contemporary drivers of change and the loss of ecological memory. *For. Ecol. Manag.* 421, 98–108. <https://doi.org/10.1016/j.foreco.2018.01.010>.
- Wiggins, G.J., Grant, J.F., Windham, M.T., Vance, R.A., Rutherford, B., Klein, R., Johnson, K., Taylor, G., 2004. Associations between causal agents of the beech bark disease complex [*Cryptococcus fagisuga* (Homoptera: Cryptococcidae) and *Nectria* spp.] in the Great Smoky Mountains National Park. *Environ. Entomol.* 33, 1274–1281. <https://doi.org/10.1603/0046-225x-33.5.1274>.
- Wiken, E., Francisco Jiménez Nava, Griffith, G., 2011. North American Terrestrial Ecoregions—Level III. In: Commission for Environmental Cooperation, Montreal, Canada.
- Witter, J.A., Stoyenhoff, J.L., Petrillo, H.A., Yocum, J.L., Cohen, J.L., 2005. Effects of beech bark disease on trees and ecosystems. In: Evans, C.A., Lucas, J.A., Twery, M.J. (Eds.), *Beech Bark Disease: Proceedings of the Beech Bark Disease Symposium*. Gen. Tech. Rep. NE-331. Newton Square PA, US. Department of Agriculture Forest Service, Northern Research Station, pp. 28–35. <https://doi.org/10.2737/NE-GTR-331>.